Activity 4 — GE Adventure Series — Reframing the Problem Instead of Fixing the Machine

Course	Fast-Track Innovations with Agile Design Thinking and Generative AI (GenAI)
WSQ Course Code	TGS-2024049781
Topic	Topic 02: Problem Framing and Ideation: Leveraging Design Thinking and Generative AI
Duration	50 minutes · Group size: 3–5 participants
Tools	Design Thinking Studio, ChatGPT or Microsoft Copilot

Learning Outcome Addressed

LO2 · A3 · A5 — Synthesise information from different sources and stakeholders to fully understand end-user needs.

The Real Case

GE Healthcare's MRI scanners were technically excellent, but designer Doug Dietz watched a small child cry with terror in front of one of his own machines. Up to 80% of paediatric patients had to be sedated to complete a scan — meaning an anaesthetist on standby, longer slots, higher cost and real clinical risk. The engineering brief would have been 'make the scanner quieter'. Dietz reframed it: the problem was not the machine's noise, it was a terrifying experience for a frightened child. The team painted the scanner rooms as pirate ships, jungles and space adventures, and trained operators to narrate the scan as an adventure — 'hold very still while the ship goes through the asteroid field'. Sedation rates dropped dramatically (reported as low as around 10% in redesigned suites) and patient satisfaction rose sharply. Not one component of the scanner's imaging technology was changed.

Your Scenario

You work for a Singapore polyclinic group. Elderly patients frequently miss or abandon their health screening appointments. The operations team's proposed solution is an SMS reminder system with a confirm button. Screening uptake has not moved in two years despite three reminder revamps.

What You Will Do

Learners practise the reframing move that turns a technical brief into a human one: building an empathy map from the case, extracting a Point of View statement, and generating How Might We questions at the right altitude — neither so broad they are useless nor so narrow they smuggle in the solution.

What you will produce

A completed Empathize and Define stage in the Design Thinking Studio: empathy map, POV statement and a voted set of How Might We questions for the polyclinic scenario.

Step-by-Step Instructions

1. Trainer clicks Create New Workspace at https://alfredang.github.io/designthinking/ and shares the workspace code or QR.
2. Click Join Workspace, enter the code and your display name.
3. Open the Empathize stage. In Empathy Map, post to Says, Thinks, Does and Feels for an elderly patient who skipped a screening.
4. Post at least two entries into User Pain Points that are NOT about forgetting the appointment.
5. Move to the Define stage. In Problem Statement, write a POV: '[User] needs [need] because [insight]'.
6. In How Might We Questions, post three HMW questions at different altitudes.
7. Use ChatGPT to generate five more HMW variants, then post only the ones that pass the altitude test.
8. Vote on the HMW questions using the voting control; the top-voted question carries into Activity 5.

Discussion Questions

1. The engineering brief was 'make the scanner quieter'. Dietz's brief was different. State both as problem statements and explain what the reframe unlocked.
2. GE changed the experience, not the technology. In the polyclinic scenario, what is the 'paint the scanner' equivalent — and what is the 'make it quieter' trap?
3. Write a How Might We that is too broad, one that is too narrow, and one that is just right. What makes the third one workable?
4. Whose perspective is missing from an empathy map built only from the operations team's data?
5. Sedation rate was the number that proved the redesign worked. What single number would prove your polyclinic solution worked?

How You Know You Are Done

Your Define stage holds a POV statement traceable to a specific empathy-map entry, and a top-voted HMW question that neither names a solution nor is too vague to answer.

Debrief — What the Room Should Conclude

Reframing is the highest-leverage move in design thinking and the one teams skip under deadline pressure. 'Make the scanner quieter' produces an incrementally better scanner; 'how might we make a frightened seven-year-old feel brave' produces a pirate ship. Note what did not change: the imaging hardware, the physics, the budget for a new machine. The constraint was the same; the problem statement was different. The altitude test for a How Might We question: if there is only one obvious answer it is too narrow (the solution is hiding inside the question); if you cannot imagine any concrete answer it is too broad. For the polyclinic, the reminder-system framing assumes the barrier is forgetting. If the real barrier is fear of a bad result, fear of cost, or having nobody to accompany them, no reminder will ever fix it — and two years of flat uptake is the evidence.